Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
PI Name:	Crum, Lawrence A. Ph.D.		
Project Title:	Guided High Intensity Focused Ultrasound (HIFU) from Mission-Critical Care		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
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Zip Code:	98105-6606	**Congressional District:**	7
Comments:			
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Bailey, Michael (University of Washington) Carter, Stephen (University of Washington) Kaczkowski, Peter (University of Washington) Vaezy, Shahram (University of Washington)		
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Task Description:

1. Specific Aims The principal objective of this NSBRI Smart Medical Systems Team project is to develop an image-guided ultrasound therapy system for mission critical care. In long-term space flight missions, a number of medical situations could develop that if not adequately addressed would result in mission failure. For example, although gravity is significantly reduced in space, inertia is not, and the collision of an astronaut with a heavy object could result in blunt internal trauma and its often associated internal bleeding. We have demonstrated that a device that produces High Intensity Focused Ultrasound (HIFU) can be combined with a device that provides ultrasound imaging to produce a duplex system that can both image a particular condition of interest and provide therapy to relieve the condition. "Image-Guided Therapy" provides enormous potential for the treatment of a variety of medical conditions.
Specific Aim 1. To develop a combined ultrasound guidance and therapy system. This first-generation system should have the following components: (A) Laptop computer control, (B) Software control, (C) Compatibility with commercial ultrasound imaging systems, (D) Single element resonant transducer, and (E) Dynamic depth focusing.

Specific Aim 2. To perform studies on the combined system that would lead to optimal performance parameters. Among the studies to be performed are the following: (A) Biological effects, tissue necrosis, acoustic hemostasis, (B) Image quality for diagnosis, (C) Targeting and monitoring capabilities, (D) Acoustic focusing and power requirements, and (E) Thermal focusing limitations.

Specific Aim 3. Utilizing the results of SAs 1 & 2, to develop an integrated ultrasound guidance and therapy system.

2. Key Findings Our 3 Specific Aims have been met. Aim 1 was completed and reported in year 1. The ability to use ultrasound to guide HIFU is the foundation of the integrated system developed in Aim 3. Aim 2 is the science to make the system clinically useful. Our group continues to lead the world in the field of acoustic hemostasis. The NSBRI project has leveraged other funding to use HIFU to stop bleeding. And we have found we can break kidney stones with HIFU. In particular the NSBRI research has led to the 2 most powerful ways to enhance or optimize tissue heating – short, strong pulses to exploit nonlinear wave propagation and dual-frequency mixing to exploit cavitation bubbles. Aim 3 has lead to the most portable and versatile guided HIFU system we know of. We have developed (A) Integrated imaging and HIFU therapy transducers [see list of references], (B) Cavitation feedback , (C) Attenuation/thermal feedback to localize treatment site, (D) Perfusion and back-scatter imaging for treatment localization, and (E) Software control and friendly user interface.

2b. Key Findings of Year 3:

First we found we could integrate the three key findings of year 2 into a useful part of our system and expand of those findings. The findings were not solely academic discoveries but became enabling technologies. One, the protocol of using short high-amplitude pulses has been integrated in the system. We showed that doubling the acoustic amplitude of the source cuts the treatment time by more than ½. In year three, we utilized this technology and defined the mechanisms underlying the effect. Two, our lesion identification algorithm has been improved to even measure temperature of HIFU heated tissue [references in list] before a lesion even forms. This enables safe and accurate targeting. Tests are underway of the system connected to the HDI-1000 imager with this imaging capability. Three, the patenting and licensing process continues for our circuit to permit real-time synchronization of HIFU therapy with an arbitrary ultrasound imager. Commercial partners may soon be using the technology to develop even smaller and more powerful systems than we have developed. Second, we developed a complete integrated guided system within a suitcase (no option for figures). The system could be synchronized with an arbitrary ultrasound imager, such as the Philips HDI-5000 on the ISS. Or we added in year three to use a simplified Doppler imaging system to locate the HIFU focus on a major bleed by visual or audio guidance. This is an uncomplicated user-friendly device to rapidly and accurately treat a mission critical bleed. Third, we discovered using 2 frequencies instead of one HIFU frequency we could accelerate hemostasis times 25%. We completed a careful study numerically and experimentally in vivo and in vitro that showed how two frequencies can increase cavitation, enhance heat deposition and accelerate hemostasis.

3. Impact The components now weigh less than 7 kg, down from more than 30 kg in year 1, and 40 kg at the start of the project. They are packaged in a single chassis (a suitcase) and operate with the Philips HDI-5000 ultrasound imager on the ISS or whatever imager is chosen for long duration missions (i.e., a Terason or a Sonosite imager such as used in the SMS project lead by Dr. J. Thomas). Doppler makes a simple rapid method stop major bleeds without even the use of the ultrasound imager. We have established a protocol to sweep short-duration, high-amplitude pulses produced by the dynamic focusing transducer to treat large areas rapidly without complications from cavitation. This includes sealing exposed lung from blood and air leakage. These findings expand our trauma care capabilities, improve therapy efficiency, and minimize the power draw required by the device. When 2 frequencies are mixed, the enhancement is even greater. Bleeds stop 25% quicker. In addition, since heating and greater, higher volume bleeds can likely be stopped. Lastly, since the heated region is broader, likely larger vessels can be treated than were previously possible.

4. Plans Detailed in our renewal proposal.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Thus, a practical flexible HIFU system for space use is nearing reality, and simultaneously, CIMU, with NSBRI funding, is one of the leaders making high intensity focused ultrasound an earth-bound clinical reality. An MRI-guided HIFU device received FDA approval for the treatment of uterine fibroids this year. The company employs a former student who graduated from our group and has made use of CIMU's published work. Devices in China have treated more than 10000 patients for cancer and are now negotiating exports to selected sites. CIMU has been consulted on these negotiations. Smaller hemostasis devices are being developed by 2 spin-off companies from CIMU. These new companies are beginning to generate the substantial private funding necessary to overcome the production, regulatory and insurance reimbursement challenges of a new therapeutic modality.

Task Progress:	Aim 1 was completed and reported in year 1. The ability to use ultrasound to guide HIFU is the foundation of the integrated system developed in Aim 3. Aim 2 is the science to make the system clinically useful. Our group continues to lead the world in the field of acoustic hemostasis. The NSBRI project has leveraged other funding to use HIFU to stop bleeding. And we have found we can break kidney stones with HIFU. In particular the NSBRI research has led to the 2 most powerful ways to enhance or optimize tissue heating – short, strong pulses to exploit nonlinear wave propagation and dual-frequency mixing to exploit cavitation bubbles. Aim 3 has lead to the most portable and versatile guided HIFU system we know of. We have developed (A) Integrated imaging and HIFU therapy transducers [see list of references], (B) Cavitation feedback , (C) Attenuation/thermal feedback to localize treatment site, (D) Perfusion and back-scatter imaging for treatment localization, and (E) Software control and friendly user interface.
Bibliography Type:	Description: (Last Updated: 03/22/2019)
Articles in Peer-reviewed Journals	Bailey MR, Pishchalnikov YA, Sapozhnikov OA, Cleveland RO, McAteer JA, Miller NA, Pishchalnikova IV, Connors BA, Crum LA, Evan AP. "Cavitation detection during shock wave lithotripsy." Ultrasound Med Biol. 2005 Sep;31(9):1245-56. PMID: 16176791 , Sep-2005
Awards	Crum LA. "Early Career Award (International Congress on Acosutics), January 2004." Jan-2004
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Dissertations and Theses	Owen NR. "An amplifier and synchronization system for a portable, ultrasound guided high intensity focused ultrasound therapy system. " Masters Thesis, University of Washington, Department of Electrical Engineering, April 2004. , Apr-2004
Patents	2031-3966DL. Patent, June 2003. Jun-2003 Owen NR, Hossack J, Bailey MR. "Method and system to synchronize acoustic therapy with ultrasound imaging."
Presentation	Bailey, M. R. "Behavior and application of cavitation in lithotripsy and high intensity focused ultrasound " Jun-2004
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Presentation	L. Crum, M. Bailey, A. Brayman, J. Guan, J. H. Hwang, V. Khokhlova, T. Matula, C. Miao, J. Tu, O. Sapozhnikov, and S. Vaezy "Recent Studies of the Physics of Ultrasound Contrast Agents and their Therapeutic Biological Applications " Jan-2004
Presentation	M. R. Bailey, J. A. Reed, A. D. Maxwell, W. Kreider, V. A. Khokhlova, V. Zderic, S. Vaezy, and L. A. Crum "Dual frequency HIFU to accelerate hemostasis " Nov-2004
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Presentation	V. A. Khokhlova, M. R. Bailey, J. A. Reed, and P. J. Kaczkowski "The relative effects of cavitation and nonlinear ultrasound propagation on HIFU lesion dynamics in a tissue phantom " May-2004